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Oligopoly

BETWEEN MONOPOLY AND PERFECT COMPETITION

- Imperfect competition refers to those market structures that fall between perfect competition and pure monopoly.
- Imperfect competition includes industries in which firms have competitors but do not face so much competition that they are price takers.

BETWEEN MONOPOLY AND PERFECT COMPETITION

- Types of Imperfectly Competitive Markets
  - *Oligopoly*
    - Only a few sellers, each offering a similar or identical product to the others.
  - *Monopolistic Competition*
    - Many firms selling products that are similar but not identical.

MARKETS WITH ONLY A FEW SELLERS

- Because of the few sellers, the key feature of oligopoly is the tension between cooperation and self-interest.
- Characteristics of an Oligopoly Market
  - Few sellers offering similar or identical products
  - Interdependent firms
  - Best off cooperating and acting like a monopolist by producing a small quantity of output and charging a price above marginal cost

A Duopoly Example

- A duopoly is an oligopoly with only two members. It is the simplest type of oligopoly.

Competition, Monopolies, and Cartels

- The duopolists may agree on a monopoly outcome.
  - *Collusion*
    - An agreement among firms in a market about quantities to produce or prices to charge.
  - *Cartel*
    - A group of firms acting in unison.

Competition, Monopolies, and Cartels

- Although oligopolists would like to form cartels and earn monopoly profits, often that is not possible. Antitrust laws prohibit explicit agreements among oligopolists as a matter of public policy.

The Equilibrium for an Oligopoly

- A *Nash equilibrium* is a situation in which economic actors interacting with one another each choose their best strategy given the strategies that all the others have chosen.

The Equilibrium for an Oligopoly

- When firms in an oligopoly individually choose production to maximize profit, they produce quantity of output greater than the level produced by monopoly and less than the level produced by competition.
- The oligopoly price is less than the monopoly price but greater than the competitive price (which equals marginal cost).

How the Size of an Oligopoly Affects the Market Outcome

- As the number of sellers in an oligopoly grows larger, an oligopolistic market looks more and more like a competitive market.
- The price approaches marginal cost, and the quantity produced approaches the socially efficient level.

GAME THEORY AND THE ECONOMICS OF COOPERATION

- *Game theory* is the study of how people behave in strategic situations.
- Strategic decisions are those in which each person, in deciding what actions to take, must consider how others might respond to that action.

GAME THEORY AND THE ECONOMICS OF COOPERATION

- Because the number of firms in an oligopolistic market is small, each firm must act strategically.
- Each firm knows that its profit depends not only on how much it produces but also on how much the other firms produce.

The Prisoners' Dilemma

- The *prisoners' dilemma* provides insight into the difficulty in maintaining cooperation.
- Often people (firms) fail to cooperate with one another even when cooperation would make them better off.

The Prisoners' Dilemma

- The prisoners' dilemma is a particular "game" between two captured prisoners that illustrates why cooperation is difficult to maintain even when it is mutually beneficial.

Figure 2 The Prisoners' Dilemma

		Bonnie' s Decision	
		Confess	**Remain Silent**
Clyde's Decision	**Confess**	Bonnie gets 8 years Clyde gets 8 years	Bonnie gets 20 years Clyde goes free
	Remain Silent	Bonnie goes free Clyde gets 20 years	Bonnie gets 1 year Clyde gets 1 year

Oligopolies as a Prisoners' Dilemma

- The dominant strategy is the best strategy for a player to follow regardless of the strategies chosen by the other players.
- Cooperation is difficult to maintain, because cooperation is not in the best interest of the individual player.

Figure 3 Jack and Jill's Oligopoly Game

		Jack's Decision	
		High Production: 40 Gal.	Low Production: 30 gal.
Jill's Decision	High Production 40 gal.	Jack gets $1,600 profit Jill gets $1,600 profit	Jack gets $1,500 profit Jill gets $2,000 profit
	Low Production 30 gal.	Jack gets $2,000 profit Jill gets $1,500 profit	Jack gets $1,800 profit Jill gets $1,800 profit

Oligopolies as a Prisoners’ Dilemma

- Self-interest makes it difficult for the oligopoly to maintain a cooperative outcome with low production, high prices, and monopoly profits.

Figure 5 A Common-Resource Game

Chevron's Decision / Exxon's Decision	Drill Two Wells	Drill One Well
Drill Two Wells	Exxon gets $4 million profit Chevron gets $4 million profit	Exxon gets $3 million profit Chevron gets $6 million profit
Drill One Well	Exxon gets $6 million profit Chevron gets $3 million profit	Exxon gets $5 million profit Chevron gets $5 million profit

Summary

- The prisoners' dilemma shows that self-interest can prevent people from maintaining cooperation, even when cooperation is in their mutual self-interest.
- The logic of the prisoners' dilemma applies in many situations, including oligopolies.